

WEEK 11 MOCK INTERVIEW MASTER SCRIPT & HIGHLIGHT REEL GUIDE

Candidate: Brian Sigei | Role: Operational Data Analyst & Analytics Engineer | Duration: 30 Minutes

Part 1: Introduction & Career Narrative (00:00 - 05:30)

Q1: 'Tell us about your background and how you built the KEMSA Supply Chain Intelligence Platform.'

Response (STAR): 'I am an Operational Data Analyst and Analytics Engineer. In Kenya's public healthcare sector, over 30% of rural clinics face critical medicine stockouts while referral hospitals suffer KES 45.2M in annual expiry losses. As the analytics and visualization lead on the Ironclad team, I engineered an end-to-end intelligence layer ingesting 4.9M records across 47 counties. We built a LightGBM 7-day risk classifier and an AI redistribution engine that cut stockouts by 68.3% and unlocked KES 39.08M in verified net annual savings with a 2.6-month payback period.'

Part 2: Deep Technical Analytics & ETL Pipeline (05:31 - 12:45)

Q2: 'How did you handle messy real-world clinic logs, missing values, and database scalability?'

Response: 'We architected a Kimball Star-Schema warehouse with 5 Dimension tables and 6 Star Fact tables in SQLite WAL mode. We automated cleaners for 7 real-world data flaws: handling negative stock via absolute reconciliation, removing future-dated logs, and deriving missing consumption using patient-demand multipliers multiplied by sub-county median visit volumes. This reduced analytical dashboard query latencies to sub-second speeds.'

Part 3: Machine Learning & Class Imbalance (12:46 - 19:15)

Q3: 'Why choose LightGBM for stockout risk, and how did you resolve class imbalance?'

Response: 'Stockouts represent a severe minority class in historical daily logs. We selected LightGBM because of its histogram-based speed, native categorical feature handling, and built-in scale_pos_weight parameter. By weighting minority stockout instances inversely to their frequency and tuning decision thresholds via F1-maximization on validation sets, our model achieved a 91.2% ROC-AUC and 88.4% Recall, proving that Days-of-Stock and supplier reliability were far stronger predictors than historical consumption lags.'

Part 4: Algorithmic Optimization & Logistics (19:16 - 25:00)

Q4: 'How does your redistribution engine ensure donor hospitals aren't left stranded?'

Response: 'The algorithm enforces a hard mathematical safety constraint: before any surplus is earmarked for transfer, the source facility must retain a minimum buffer of (Safety Stock + Supplier Lead Time Days) consumption. It then pairs surplus hubs (>60 DOS) with nearby deficit clinics (<7 DOS) within a 400km radius, optimizing Haversine transport distance and cutting emergency replenishment costs by 65.5%.'

Part 5: CFO Business Case Defense & Close (25:01 - 30:00)

Q5: 'Walk the Board through your financial math. Why should we approve KES 8.5M today?'

Response: 'Our solution achieves KES 41.88M in gross annual savings (KES 19.03M from reduced expiry and KES 18.60M from eliminated emergency surcharges). Subtracting annual cloud opex of KES 2.80M leaves a net annual cash flow of KES 39.08M. Dividing our KES 8.50M Capex by KES 39.08M yields a Payback Period of exactly 2.61 months (~78 days). Year 1 net ROI is 359.8%, with a 3-Year NPV of KES 85.34M. This is a non-disruptive, self-funding investment that transforms public health delivery across Kenya.'